Diata Rodriguez

9.5

CHE 313 Transport Phenomena III
Homework #2 (Due:9/25/25, 1:00 pm)
TA office hour (T/TR, 3-5 pm): Aelx Acquah (aacqua2@uic.edu)

<u>Note: Explain your answers in detail!</u>

1. Textbook (Wankat, P. C. 5th edition), 2.D5.

2. Textbook (Wankat, P. C. 5th edition), 2.D10.

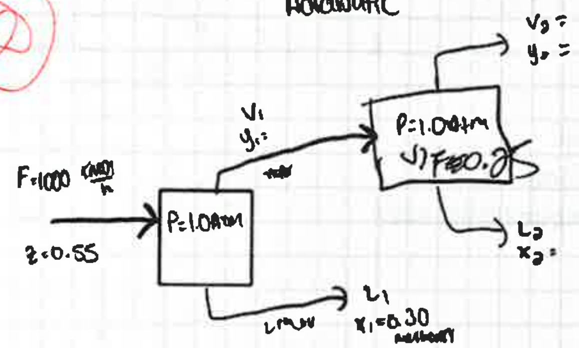
3. Textbook (Wankat, P. C. 5th edition), 2.D16.

4. Textbook (Wankat, P. C. 5th edition), 2.D24. Please use a spreadsheet and submit it with your answer and graph.

5. Textbook (Wankat, P. C. 5th edition), 3.D3.

2.05 Feed that is a binary mixture of methanol & water

Adiabatic 55mol% 45mol%



a) Find y_1, T_1 (y_1) & vapor flow rate

$F = L + V$

$Fz = Lx + Vy$

$x = 0.3$

$y = -\frac{L}{V}x + \frac{F}{V}z$

methanol

$y_1 = 0.665$

$T_1 = 78.0°C$

$y = -\frac{L}{V}(0.3) + \frac{F}{V}(z)$

$Fz = (1-\frac{V}{F})Fx + \frac{V}{F}Fy$

$z = (1-\frac{V}{F})x + y$

$0.55 = (1-\frac{V}{F})(0.3) + (0.665)\frac{V}{F}$

$x = (1-\frac{V}{F})F$

$0.55 = 0.3 - 0.3\frac{V}{F} + 0.665\frac{V}{F}$

$V = \frac{V}{F}(F)$

$0.55 = 0.3 + 0.365\frac{V}{F}$

$0.25 = 0.365\frac{V}{F}$

$\frac{V}{F} = 0.685$

$x = (1-0.685)1000 = 315$ kmol/h

vapor flow rate $y_1 = (0.685)1000 = 685 \frac{kmol}{h}$

b) Second drum: y_2, x_2, T_2, vapor flow rate y_2

$\frac{V}{F} = 0.25$

$F_2 = V_1 = 685 \frac{kmol}{hr}$ $y_1 = z_1$

$V_2 = 0.25(685 \frac{kmol}{hr}) = 171.3 \frac{kmol}{h}$

$x = 0.60$ $y = 0.825$ $T = 71.2°C$

$x = 0.70$ $y = 0.870$ $T = 69.3°C$

$z = (1-\frac{V}{F})x + \frac{V}{F}y$

$0.665 = (0.75)x + (0.25)y$

T_2

$\frac{69.3 - 71.2}{0.7 - 0.6} = \frac{-1.9}{0.1}$

$y = -\frac{L}{V}x + \frac{F}{V}z$

F = 1

L = 0.75

V = 0.25

$y_2 = 0.83$

$x_2 = 0.61$

$y = -3x + 2.66$

$(y_2 - y_1)$

$$y = 71.2 + (x - 0.60)\frac{(69.3 - 71.2)}{(0.70 - 0.60)}$$

($x_2 - x_1$ written above $(x - 0.60)$)

$x = 0.61$

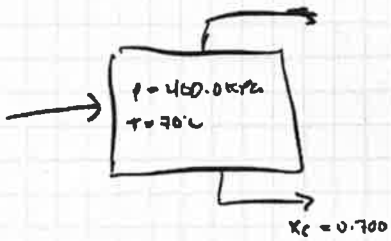
$T_2 = 71$

D10) 2

z C4 n-butane = 0.35

z C3 propane =

z C6 n-hexane =



find the mol fraction of n-hexane in feed

z_{C6}? & V/F

K values

$\cancel{K_{C4} = 1.98}$
$\cancel{K_{C3} = 3.25}$
$\cancel{K_{C6} = 0.13}$

$K_{C3} = 5$
$K_{C4} = 2$
$K_{C6} = 0.3$

$$X_i = \frac{z_i}{1 + \frac{V}{F}(K_i - 1)}$$ eq from notes

$$X_3 = 0.700 = \frac{z_{C6}}{1 + \frac{V}{F}(0.3 - 1)}$$

$$\frac{z_{C6}}{1 + 0.3\frac{V}{F} - \frac{V}{F}}$$

$$z_{C6} = 0.700\left(1 - 0.7\frac{V}{F}\right)$$

$$z_{C3} = 1 - z_{C4} - z_{C6}$$

$$z_{C3} = 1 - 0.35 - z_{C6}$$

$$z_{C3} = 0.65 - 0.700\left(1 - 0.7\frac{V}{F}\right)$$

$$z_{C3} = 0.65 - 0.700 - 0.49\frac{V}{F}$$

$$z_{C3} = -0.05 - 0.49\frac{V}{F}$$

Rachford - Rice

$$\sum \frac{(K_i - 1) z_i}{1 + (K_i - 1) V/F}$$

$$\frac{z_{C3}(5.0 - 1)}{1 + (5.0 - 1)\frac{V}{F}} + \frac{0.35(2 - 1)}{1 + (2 - 1)\frac{V}{F}} + \frac{z_{C6}(0.3 - 1)}{1 + (0.3 - 1)\frac{V}{F}}$$

$$z_{C6} = 0.700\left(1 - 0.7\frac{V}{F}\right)$$

$$\frac{z_{C3}(4.0)}{1 + 4\frac{V}{F}} + \frac{0.35}{1 + \frac{V}{F}} + \frac{0.700\left(1 - 0.7\frac{V}{F}\right)(-0.7)}{1 - 0.7\frac{V}{F}}$$

$$0.700(-0.7) = -0.490$$

$$\frac{\left(-0.050 + 0.490\frac{V}{F}\right)(4.0)}{1 + 4\frac{V}{F}} + \frac{0.35}{1 + \frac{V}{F}} - 0.490 = 0$$

0.43

$\frac{V}{F} = 0.1$

$$\frac{-0.050 + 0.490(0.1)(4.0)}{1 + 4(0.1)} + \frac{0.35}{1 + (0.1)} - 0.490$$

$$\frac{-0.050 + 0.196}{1 + 0.4} + \frac{0.35}{1.1} - 0.490$$

$$= 0.104 + 0.318 - 0.490$$

$$= -0.068$$

$\frac{V}{F} = 0.2$

$$\frac{4(-0.050 + 0.098)}{1.8} + \frac{0.35}{1.2} - 0.490$$

$$0.107 + 0.292 - 0.490 = -0.091$$

$\frac{V}{F} = 0.45$

$$\frac{(-0.050 + 0.221)}{2.8} + \frac{0.350}{1.45} - 0.490$$

$$0.244 + 0.241 - 0.490$$

$\frac{V}{F} = 0.46$

$$\frac{4(-0.05+0.225)}{2.84} + \frac{0.350}{1.46} - 0.490$$

converges

$= 0.246 + 0.240 - 0.490 = -0.004$

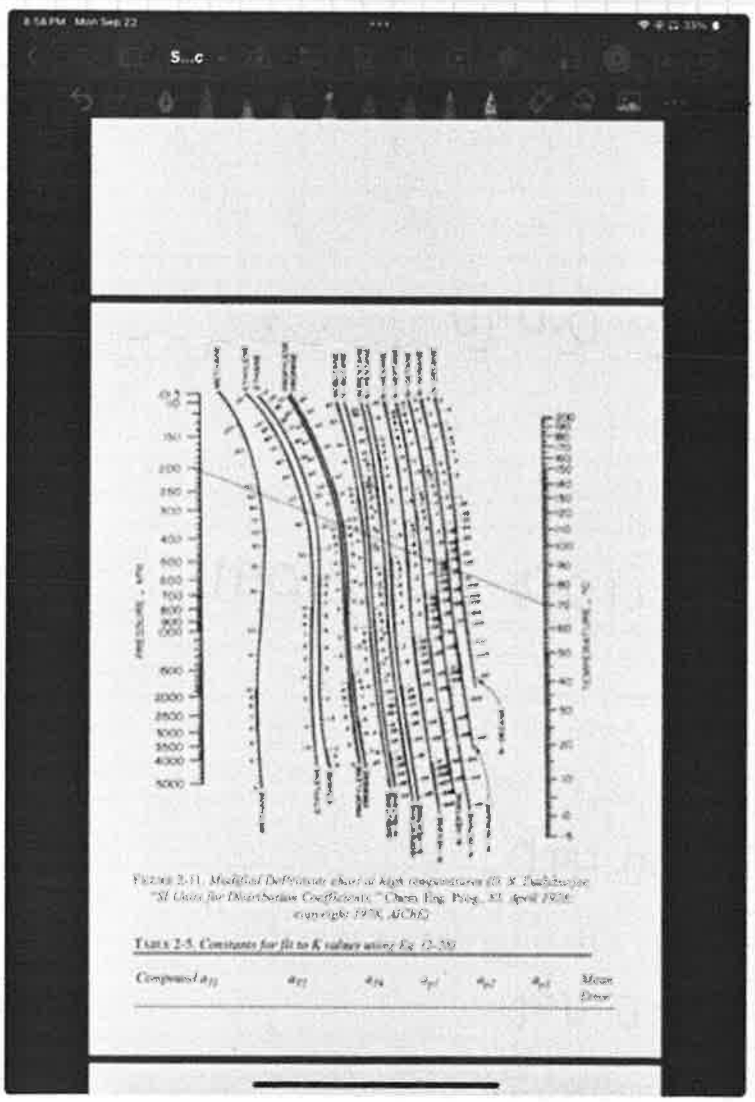
$\frac{V}{F} = 0.46$

find z_{C6} using $\frac{V}{F} = 0.46$

$z_{C6} = 0.700(1 - 0.7\frac{V}{F})$

$z_{C6} = 0.700(1 - 0.7(0.46))$

$= 0.700(1 - 0.322) = 0.47$



range

$\frac{V}{F} = 0.46 - 0.48$

>16

$F = 150.0\ \frac{kmol}{h} \rightarrow$ [$T_{drum} = 10°C$, $P_{drum} = 250\ kPa$] $\rightarrow V, y$; $\rightarrow L, x$

$z_{C1} = 0.5$
$z_{C4} = 0.1$
$z_{C5} = 0.15$
$z_{C6} = 0.25$

Rachford Rice

$$\sum_{i=1}^{4} \frac{(K_i - 1) z_i}{1 + (K_i - 1)\frac{V}{F}}$$

$K_{C} = 80$
$K_{C4} = 0.57$
$K_{C5} = 0.16$
$K_{C6} = 0.06$

$$\frac{(80-1)0.5}{1+(80-1)\frac{V}{F}} + \frac{(0.57-1)0.1}{1+(0.57-1)\frac{V}{F}} + \frac{(0.16-1)0.15}{1+(0.16-1)\frac{V}{F}} + \frac{(0.06-1)0.25}{1+(0.06-1)\frac{V}{F}}$$

$$\frac{39.5}{1+79\frac{V}{F}} + \frac{-0.043}{1-0.43\frac{V}{F}} + \frac{-0.126}{1-0.84\frac{V}{F}} + \frac{-0.235}{1-0.94\frac{V}{F}}$$

Newtonian

$$\frac{df_K}{d(\frac{V}{F})} = -\sum \frac{(K_i - 1)^2 z_i}{[1 + (K_i - 1)\frac{V}{F}]^2} = -\left[\frac{(80-1)^2 0.5}{[1+(80-1)\frac{V}{F}]^2} + \frac{(0.57-1)^2 0.1}{[1+(0.57-1)V/F]^2} + \frac{(0.16-1)^2 0.15}{[1+(0.16-1)\frac{V}{F}]^2} + \frac{(0.06-1)^2 0.25}{[1+(0.06-1)\frac{V}{F}]^2}\right]$$

$$= -\left[\frac{3120.5}{1+(79\frac{V}{F})^2} + \frac{0.01849}{(1-0.43\frac{V}{F})^2} + \frac{0.10584}{(1-0.84\frac{V}{F})^2} + \frac{0.2209}{(1-0.94\frac{V}{F})^2}\right]$$

$\left(\frac{V}{F}\right) = 0.3 \quad \frac{df_K}{d(\frac{V}{F})} = -5.76 \quad RR = 1.05 \neq 0$

for a new $\frac{V}{F}$

$$\frac{V}{F} = \left(\frac{V}{F}\right)_c - \frac{f_K}{\frac{df_K}{d(\frac{V}{F})}} = 0.3 + \frac{1.05}{5.76} = 0.48$$

$\frac{V}{F} = 0.48$ Newtonian $= -3.12$ $RR = 0.32$

$$= 0.48 - \frac{0.32}{-3.12} \neq 0.58$$

$\frac{V}{F} = 0.58$ Newtonian $= -2.93$ $RR = 0.024$

$= 0.58 + \frac{0.024}{2.93} = 0.588$

(4) $0.588 = \frac{V}{F}$

$RR = 4.98 \times 10^{-4}$ converses < 0

$\frac{V}{F} = 0.588$

$x_i = \frac{z_i}{1+(K_i-1)\frac{V}{F}}$ $y_i = K_i x_i$

C_1 : $x = 0.011$ $y = 0.88$

C_4 : $x = 0.134$ $y = 0.08$

C_5 : $x = 0.296$ $y = 0.05$

C_6 : $x = 0.559$ $y = 0.03$

$\frac{V}{F} = 0.588$ $V = 0.588F$

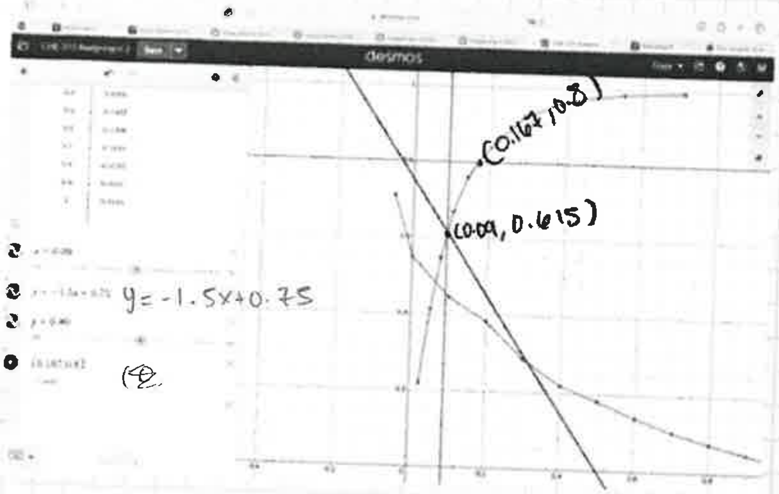
$V = 0.588(150)$

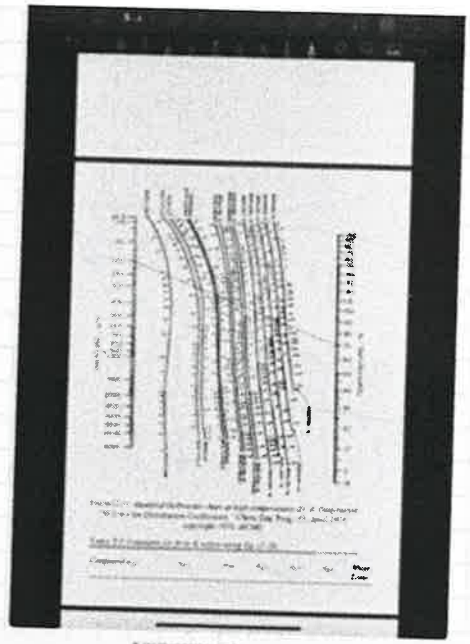
$F = 150 \frac{kmol}{h}$ $V = 88.2 \frac{kmol}{h}$

$F = V + L$

$150 \frac{kmol}{hr} = 88.2 \frac{kmol}{h} + L$

D24) Propane & Hexane





b) $z = 0.30$

$\frac{V}{F} = 0.40$

use lever rule

$z = x(1-\frac{V}{F}) + y(\frac{V}{F})$

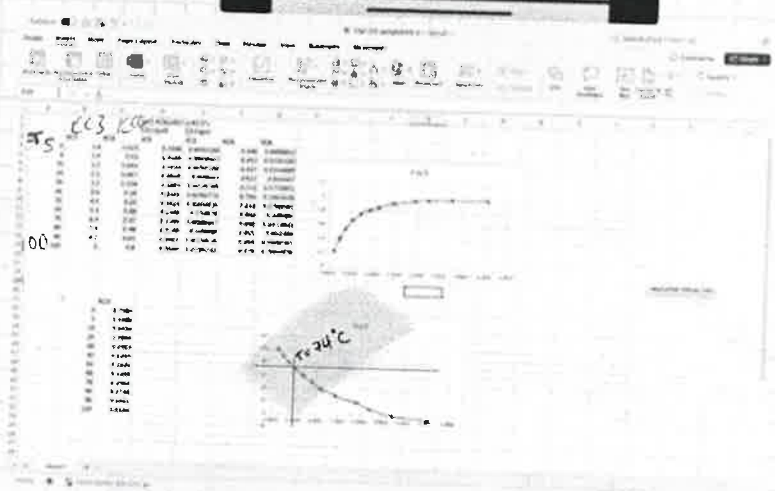
$0.30 = x(1-0.4) + y(0.4)$

$0.3 = 0.6x + 0.4y$

$y = \frac{-0.6}{0.4}x + \frac{0.3}{0.4}$

$y = -1.5x + 0.75$

crosses Data @ $(0.09, 0.615)$



C)

$y = 0.80$

$z = 0.60$

when $y_{c3} = 0.80$

$x_{c3} = 0.167$

$T = 74°C$

$z = (1-\frac{V}{F})x + \frac{V}{F}y$

$\frac{V}{F} = \frac{z-x}{y-x} = \frac{0.60-0.167}{0.80-0.167} = \frac{0.44}{0.64} =$

$\frac{V}{F} = 0.687$

D) Rachford-Rice

B) Part B

$x_{c3} = 0.09$ $y_{c3} = 0.615$

$z_{c3} = 0.30$

$\frac{V}{F} = 0.40$

$z_{c6} = 0.70$

propane: from excel

$y_{c3} = K_{c3} x$

$\frac{y_{c3}}{x_{c3}} = K_{c3} = \frac{0.615}{0.09} = 6.83$

← Tests x & y values if 0.40 converges to 0

RACHFORD RICE

$$\sum \frac{(K_i - 1) z_i}{1 + (K_i - 1) V/F}$$

$$= \frac{z_{C3}(K_{C3} - 1)}{1 + (K_{C3} - 1)\frac{V}{F}} + \frac{z_{C6}(K_{C6} - 1)}{1 + (K_{C6} - 1)\frac{V}{F}}$$

$$= \frac{0.30(5.83)}{1 + (5.83)\frac{V}{F}} + \frac{0.70(-0.577)}{1 + (-0.577)\frac{V}{F}}$$

INPUT $\frac{V}{F} = 0.40$

$$= \frac{1.749}{1 + 2.332} - \frac{0.404}{1 - 0.231} = \frac{1.749}{3.332} - \frac{0.404}{0.769} = 0$$

with our values Rachford Rice is satisfied w/ $\frac{V}{F} = 0.40$

Part D

C) Rachford Rice

calculate K values

$y_{C3} = 0.80$
$x_{C3} = 0.167$

$z_{C3} = 0.60$

$$K_{C3} = \frac{y_{C3}}{x_{C3}}$$

$y_{C6} = 0.20$

$z_{C6} = 0.40$

$x_{C6} = 0.833$

$$K_{C3} = \frac{0.80}{0.167}$$

$K_{C6} = 0.24$

$K_{C3} = 4.79$

$$\sum \frac{(K_i - 1) z_i}{1 + (K_i - 1) V/F}$$

$$\frac{(K_{C3} - 1) z_{C3}}{1 + (K_{C3} - 1)\frac{V}{F}} + \frac{(K_{C6} - 1) z_{C6}}{1 + (K_{C6} - 1)\frac{V}{F}}$$

$$\frac{(3.79)0.60}{1 + (3.79)\frac{V}{F}} + \frac{(-0.76)0.40}{1 + (-0.76)\frac{V}{F}}$$

$$\frac{(3.79)0.60}{1+(3.79(0.687))} + \frac{(-0.76)0.40}{1+(-0.76(0.687))} \qquad \frac{V}{F} = 0.687$$

$$\frac{2.274}{3.6} + \frac{-0.304}{1-0.522}$$

✓

$$0.63 \quad -0.63 = 0$$

3D3 ($\cancel{2}$ .5)

Find external boilup rate

0 = in − out

D + B = F + C

.80D + 0.05B = 0.40F + 1.0C

Energy Balance

$\dot{m} H_{in} - \dot{m} H_{out}$ + $\cancel{Q}$ + $\cancel{W}$

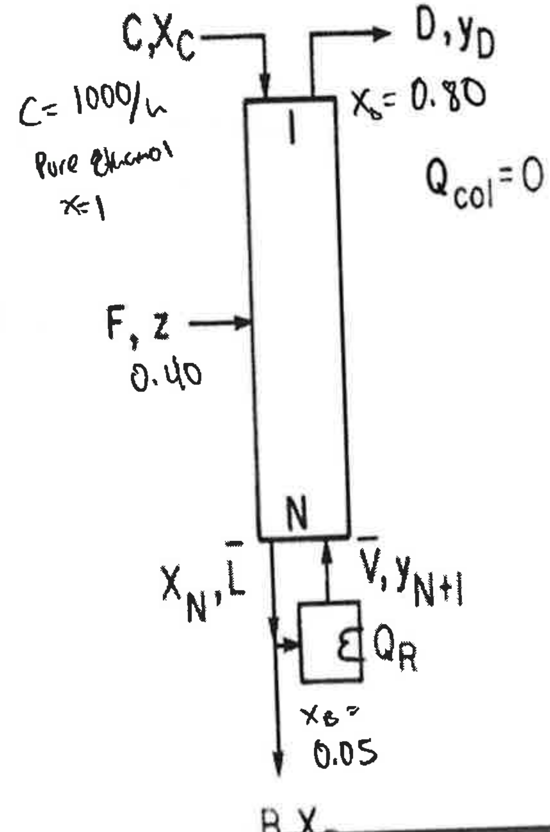
$\dot{m} H_{in} - \dot{m} H_{out} = 0$

$F h_F + C h_C + V h_V = D h_D + B h_B$

$B = F + C - D$

$F h_F + C h_C + V h_V = D h_D + F h_B + C h_B - D h_B$

$V h_V = D(h_D - h_B) + F(h_B - h_F) + C(h_B - h_C)$



$$= \frac{D(h_D - h_B) - F(h_B - h_F) - C(h_B - h_C)}{h_V}$$

$$V = \frac{D h_D + B h_B - F h_F - C h_C}{h_V}$$

subcooled liq

$h_F = -20$

$h_C = 60$

sat liq

sat vapor

$h_D = 340$

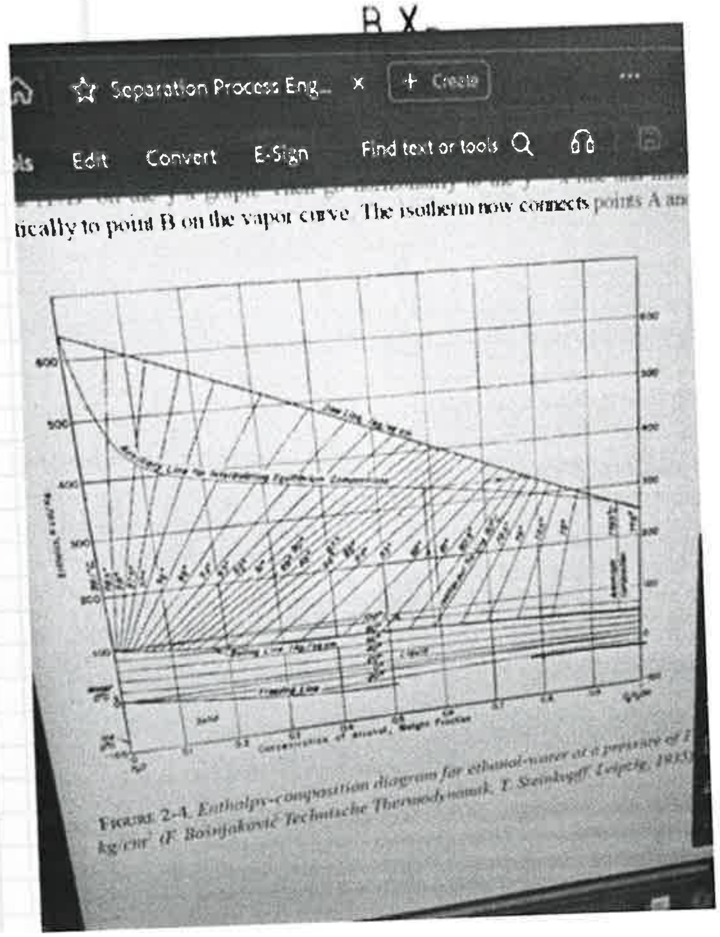
y = 0.80

$h_B = 98$ sat liq

x = 0.05

$h_V = 490$

? linear extrapolating



$$V = \frac{Dh_D + Bh_B - Fh_F - Ch_C}{h_V}$$

$D + B = F + C \quad B = F + C - D$

$0.80D + 0.05B = 0.40F + C$

$$0.80D + 0.05(F + C - D) = 0.40F + C$$

$$0.80D + 0.05F + 0.05C - 0.05D = 0.40F + C$$

$$0.75D = 0.35F + 0.95C$$

$$D = \frac{0.35F + 0.95C}{0.75}$$

Plug into energy Balance

subcooled liq

$h_F = -20$

$h_C = 60$

sat liq

sat vapor

$h_D = 340$

$y = 0.80$

$h_B = 98$ sat liq

$x = 0.05$

$h_V = 490$

sat vapor enthalpy

$x = 0.05$, $y = 0.4$

$$V = \frac{Dh_D + Bh_B - Fh_F - Ch_C}{h_V}$$

$$V = \frac{D(h_D - h_B) - F(h_B - h_F) - C(h_B - h_C)}{h}$$

$$D = \frac{0.35F}{0.75} + \frac{0.95(1000)}{0.75}$$

$$D = 0.466F + 1266.7$$

$$V = \frac{D(242) + F(118) + 1000(38)}{490}$$

.7

$$V = \frac{(0.466F + 1266.7)242 + 118F + 38000}{490}$$

$$V = \frac{112.94F + 306533 + 118F + 38000}{490}$$

$$V = 230.94F + 344541$$

$$V = \frac{230.94F + 344533}{490}$$

$$V = 0.4715F + 703.1$$

$V = 0.5969F + 899$

?

1532.4

BACK OF BOOK SOLUTION

$$F = 1760$$

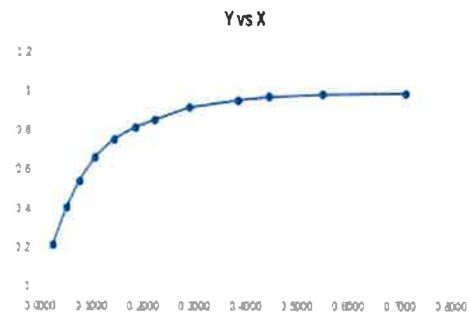
$$V = 1532.1$$

Question 4 graph

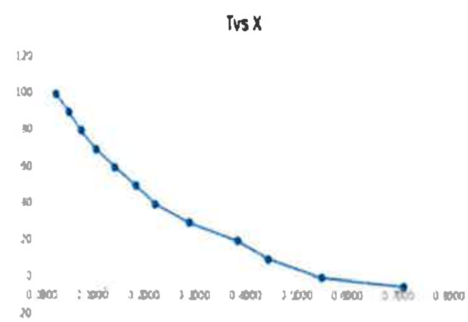
Che 313 assignment 2 – Saved

K19

			x=(1-KC6)/(KC:	y=KC3*x		
			C3 Liquid	C3 Vapor		
T	KC3	KC6	xC3	YC3	XC6	YC6
-5	1.4	0.023	0.7095	0.9933188	0.290	0.0066812
0	1.8	0.03	0.5480	0.9864407	0.452	0.0135593
10	2.2	0.044	0.4434	0.9755102	0.557	0.0244898
20	2.5	0.067	0.3835	0.958693	0.617	0.041307
30	3.2	0.109	0.2883	0.9224199	0.712	0.0775801
40	3.9	0.18	0.2204	0.8596774	0.780	0.1403226
50	4.5	0.22	0.1822	0.8200935	0.818	0.1799065
60	5.4	0.28	0.1406	0.759375	0.859	0.240625
70	6.4	0.37	0.1045	0.6686567	0.896	0.3313433
80	7.3	0.49	0.0749	0.546696	0.925	0.453304
90	8.2	0.62	0.0501	0.4110818	0.950	0.5889182
100	9	0.8	0.0244	0.2195122	0.976	0.7804878



T	XC3
-5	0.7095
0	0.5480
10	0.4434
20	0.3835
30	0.2883
40	0.2204
50	0.1822
60	0.1406
70	0.1045
80	0.0749
90	0.0501
100	0.0244



Sheet1

Ready